

# *CMS Development Standards & Best Practices*



**GOA ELECTRONICS LIMITED**

A Subsidiary of EDC Ltd. (A Government of Goa Undertaking)

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## REVISION HISTORY

| Revision No.<br>/ Version<br>No. | Date          | Prepared by            | Approved by          | Details of Change                                                                                         |
|----------------------------------|---------------|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| 1.0                              | 17-April-2023 | Sidhesh Shinde<br>(SE) | Anant Yende<br>(CTO) | Updation of security standards                                                                            |
| 1.1                              | 24-April-2024 | Sidhesh Shinde<br>(SE) | Anant Yende<br>(CTO) | Addition of standards for usage of CDNs                                                                   |
| 1.2                              | 19-Sep-2024   | Sidhesh Shinde<br>(SE) | Anant Yende<br>(CTO) | Addition of GIGW 3.0 reference URL, and addition of GIGW references to various sections in this standards |

**Audience:**

CTO, TL, SE / TSE , Testers, PM's

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**\*Please Note**, Reference URL Guidelines for Indian Government Websites (GIGW) 3.0:<https://cdnbbsr.s3waas.gov.in/s3c92a10324374fac681719d63979d00fe/uploads/2024/02/2024022987.pdf>



## II. Project Related Information & Definition:

The first step is to determine the goal of the website, considering the following parameters:

- Primary aim of the website.
  - What does the organization (who has asked for the website) do.?
  - What Problem is this website going to solve?
  - Who is the target audience?
- 

## III. Website Design:

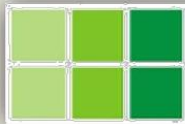
Depending on the CMS platform chosen for the website (Drupal / WordPress, etc), research should be done to find an appropriate design template for the website or create one.

The design template will act as a basis of design for the website. Either choose from existing sources that offer paid templates or get them designed by the in-house team.

Every website designed/developed for government of Goa should contain and adhere to the standard design and layout so as to ensure that citizens are able to locate content uniformly on all the government websites.

Standard website landing page layout is mentioned below for reference.:

Reference URL: <https://dpg.goa.gov.in>



# GOA ELECTRONICS LIMITED

A Subsidiary of EDC Ltd. (A Government of Goa Undertaking)

CMS Development  
Standards & Best  
Practices

Helpline number: 9319 334 335. Timings (Monday to Friday and closed on public holidays): 9:30AM to 5:45PM. Lunch break: 1:00PM to 2:00PM

GOVERNMENT OF GOA  
DIRECTORATE OF PUBLIC GRIEVANCES



WELCOME TO DIRECTORATE OF PUBLIC GRIEVANCES, GOA

Hon'ble Chief Minister  
**Dr. Pramod Sawant**

The success of a government and democracy at large is deeply rooted in its ability to listen and redress grievances of its citizens efficiently. We as a state have been taking a number of measures to improve our machinery for prompt and speedy disposal of citizen's requests and redress their grievances. The launch of this website is a major milestone in this journey which will help the government to connect with our "Goenikars" listen to their grievances and resolve them promptly.



Register Grievance

To Register your Grievance, click on the link below

[Click Here](#)

View Grievance Status

To check your Grievance status against your ACK No., click on the link below

[Click Here](#)

Grievance Statistics

To view the Public Grievance Statistics click the link below

[Click Here](#)

## EVENTS



Launch of the Public Grievance Official website on the occasion of completion of 100 Days of the Government.



Launch of the Public Grievance Official website...  
On 6th July, Hon'ble CM, Dr. Pramod Sawant on the ...  
Addressing the grievances and issues of the ...  
Hon'ble chief minister Dr. Pramod Sawant, addressing the grievances and ...  
Hon'ble Chief Minister Dr. Pramod Sawant int...  
Hon'ble Chief Minister Dr. Pramod Sawant on Sunday organized a ...



### 1. Header and Banner Section

### 2. Website Content

### 3. Important Links section

### 4. Website Footer section



For more details, please refer to: GIGW 3.0 Section: 5.1.18

Each of the section from the design referenced above are depicted detailed below:

## 1. Header and Banner Section:

Given below in a standard website template for the header of the website, that is divided into seven sections. The details of each sections are mentioned below.



### Section Details:

- 1) "1". Contains Links to state government portal and Central government portal..
- 2) "2". Accessibility features. This will include the accessibility features like:
  - a) Screen Reader Access
  - b) Skip to main content
  - c) Search Feature
  - d) Font Re-sizer
  - e) Website Site-map link.
- 3) "3". Will include the website logo and the department name whom the website belongs.
- 4) "4". Will include the logos and the links to the Government of India websites which are:
  - a) Make in India website
  - b) 150 years of Celebrating the Mahatma, website
  - c) Azadi ka Amrit Mahotsav, website.
- 5) "5". Website navigation menu only.
- 6) "6". Website banner / image slider / video.
- 7) "7". More accessibility section.



## 2. Website Content

All the content / highlights should be shown in this section, this can consist of text, hyperlinks, buttons, images, videos, etc.

## 3. Important Links.

Section consists of links to important government website in a scrolling manner.

Following are widely used important links:

- Digilocker
- Digital India
- India.gov.in
- Swachh Bharat, etc.

Quick Tip:  
Refer External Link Implementation Standards.

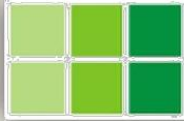
## 4. Footer Section

Given below in a standard website template for the Footer of the website, that is divided into seven sections. The details of each sections are mentioned below.



Section Details:

- 1) "1". Include the following links:
  - a) Website Disclaimer
  - b) Website Policies, which are:
    - i. Copyright Policy



- ii. Hyperlink Policy
- iii. Privacy Policy
- iv. Terms and conditions.
- v. Refund Policy (If any payment services are integrated)

*For more details, please refer to: GIGW 3.0 Sections: 5.3.3 & 5.4.3*

c) Website Site-map

*For more details, please refer to: GIGW 3.0 Section: 5.1.18*

2) “2”. Is the main website footer section. That will include more sections within it.

- a) “3”. Is a website visitor counter.
- b) “4”. contains the information about the owner and the developers of the website. It also contains the Google Language selector and the website last updated date.

*For more details, please refer to: GIGW 3.0 Sections: 5.1.12 & 5.4.10 & 5.1.5*

- c) “5” contains more important links of the Government of Goa and India websites and application

3) “6”. Contains the website copyright message in the owners name of the website.

*For more details, please refer to: GIGW 3.0 Section: 5.1.4*

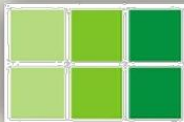
---

### III. List of standards applicable to any websites are as below:

- 1) Hosting Standards
- 2) Domain Name Standards
- 3) Technology Standards
- 4) Browser/Device Compatibility Standards
  - a) Responsive Layout
  - b) W3C Standards
- 5) Multimedia Standards
- 6) Accessibility Standards
- 7) Guidelines for Indian Government Websites (GIGW 3.0) Compatible.
- 8) API Integration
- 9) External Link Implementation standards
- 10) CDN Usage Standards.
- 11) Security Standards
- 12) Performance Standards
- 13) Search Engine Optimization (SEO) Standards
- 14) Website Management Standards

#### 1) Hosting Standards:





All the government and the public sector undertaking websites need to be hosted primarily on the NIC cloud server or at State Data Center.

Websites can also be hosted on the the public cloud servers such as Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure etc. However the cloud server needs to be located in India and should be registered with Ministry of Electronics & Information Technology (MeiTY).

All websites should be configured with SSL and should not allow access to the website without SSL. The directory write permission should be disabled on the server and only folder mentioned in the audit certificate can have write access.

*For more details, please refer to: GIGW 3.0 Section: 5.3.2*

## 2) Domain Name Standards:

Government department domain name should consists of 'gov.in' or 'nic.in' domain names. The respective department can also have 'goa.gov.in'

Alternatively Goa can also be added to the primary domain name for example: 'goadmng.gov.in' and 'dmg.goa.gov.in' are valid names

Below are examples of valid domain names:

goadmng.gov.in

goaonline.gov.in

dmg.goa.gov.in

services.goaonline.gov.in

ceogoa.nic.in

ahvs.goa.gov.in

Some examples of invalid domain names are:

goa.forest.gov.in

Ahvsgoagov.in

ahvs.goagov.in

goa.dmg.nic.in

Government corporations can consists of domain names as : .com .co.in or .in

*For more details, please refer to: GIGW 3.0 Section: 5.1.21*

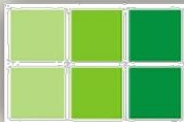
## 3) Technology Standards:

The website needs to be primarily designed using open source CMS tools such as Drupal or WordPress with the latest stable version.

The website can also be developed using other technologies for client with CTO approval.

The developer should use latest version of tools for development.

The website should use latest versions of every components used in the website such as



- JQuery (if used)
- Bootstrap (if used)
- Plugins
- PHP
- MySQL
- Drupal / WordPress /etc.
- Web server such as: XAMPP / WAMPP / IIS / Apache etc.
- Servers: Windows, Linux, CentOS, etc

#### 4) Browser / Device Compatibility Standards:

The websites being developed should be compatible with the modern browsers such as Microsoft Edge, Google Chrome, Firefox, safari, etc, and similarly mobile browsers too.

**I.) Responsive Layout:** The website and all its pages, controls, images and any media should be responsive to the web browser, screen size / device and orientation.

Quick Tip: How to ensure responsive layout:

**Setting The Viewport:** Following code should be put inside the head tag of the website.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

**Responsive Images:** Responsive images are images that scale nicely to fit any browser size. If the CSS 'width' property is set to 100%, the image will be responsive and scale up and down.

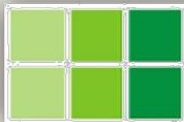
```

```

**Use of Media Queries:** With media queries you can define completely different styles for different browser sizes.

```
/* Use a media query to add a breakpoint at 800px: */
@media screen and (max-width: 800px) {
  .left, .main, .right {
    width: 100%;/*The width is 100%, when the viewport is 800px or
    smaller*/
  }
}
```

**II.) W3C Compliance:** W3C compliance is a set of criteria for HTML & CSS which ensures that the code is presented to a uniform specification as laid out by the world



wide web consortium (W3C). The web pages in the website should be complied with the W3C standards.

To validate the same: <https://validator.w3.org/> can be used to see exception and then complied.

Tools such as <https://validator.w3.org> will show all the errors found on each of the pages, highlighting what needs to be fixed in order for the page to validate.

*For more details, please refer to: GIGW 3.0 Section: 5.2*

**III.) CSS Compliance:** The web pages should use external CSS file instead of on-page / inline CSS. On-page / inline CSS can be only implemented to override certain attributes. The CSS file need to be local and cannot to referenced from external websites. The CSS file should define theme for the website. The color set should be uniform. The font should be uniform and reference from 3 fonts to be used on the website. Standard color code for button to be used in line with the theme.

By default use following font family of “Titillium Web”, Poppins, sans-serif, Roboto, Open Sans.

*For more details, please refer to: GIGW 3.0 Sections: 5.1.15 & 5.2.17 & 5.2.19*

## 5) Multimedia Data Standards

- Image format allowed: JPG, JPEG, PNG, Webp (Only if compatible with web browsers).
- Image size: Image size should not exceed 200KB.
- Video Formats allowed: MP4,
- Video size: Videos should be optimized as much as possible before using it on the website.
- In-case the government organization has its official YouTube channel, video links from the channel can be embedded on the website, instead of uploading actual video file

## 6) Accessibility Standards:

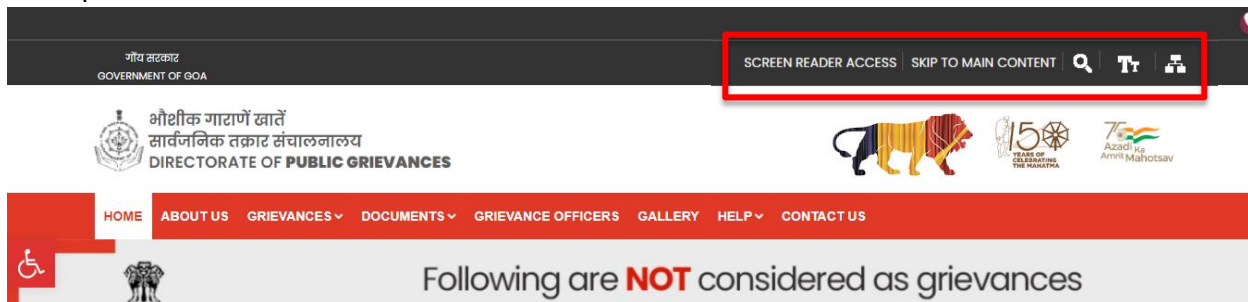
Each and every government website must have accessibility tools and function for ensuring everyone is able to use the website. The website header should prominently contain the following components:

- i. Font Size changer
- ii. Website Sitemap
- iii. Skip link to main content
- iv. Screen Reader
- v. Contrast color change (for the visually impaired).

Example 1:



Example 2:



For more details, please refer to: GIGW 3.0 Section: 5.2

## 7) Guidelines For Indian Government Websites (GIGWA) Compatible:

Guidelines for Indian Government Websites (GIGW), developed by National Informatics Centre (NIC) are mandatory to be followed throughout the life-cycle of a Government website, web portal/application right from its conceptualization to design, development, maintenance and management.

While designing the website it must be ensured that the website is GIGW complaint.

These guidelines demand a specific standards to be maintained while creating a government website.

**GIGWA Compliance Matrix:** GIGW Compliance matrix is a checklist mandatory guidelines, use this checklist to validate the websites against these guidelines and make necessary modification to ensure compliance.

The compliance matrix can be downloaded from the following link:

<https://cdnbbsr.s3waas.gov.in/s3c92a10324374fac681719d63979d00fe/uploads/2020/03/20200326100.pdf>

For more info visit: [https://darpg.gov.in/sites/default/files/gigw-manual\\_Revised2018\\_0.pdf](https://darpg.gov.in/sites/default/files/gigw-manual_Revised2018_0.pdf)

## 8) API Integration Standards:

The APIs are used to read data / content programmatically from a different source. When consuming APIs following points to be followed:

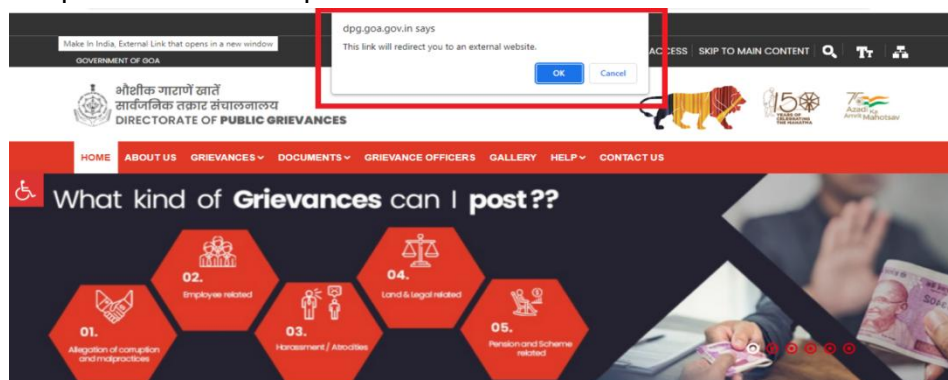
- API should be accessible over SSL only
- If possible, APIs communication should be encrypted
- The API should be whitelisted.
- Authentication methods are used if available
- Page where APIs are called, captcha to be implemented.

For more details, please refer to: GIGW 3.0 Section: 5.2.3



## 9) External Link Implementation:

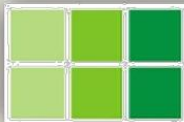
When ever there is a request made by the user by clicking an external website link on the website, the user must be presented with a prompt box asking for confirmation whether the user wants to proceed with his/her action (Example shown below). If yes, then the requested link must open in a new window



## 10) Document Upload Standards

If the website displays any documents then the same should be following below file standards:

- i. Allowed document formats: PDF, (*for image files, HTML, or files of any other format if necessary to upload then approval should be taken from the CMS Lead before upload*).
- ii. Document type should be clearly stated along with the title, size, format, expiry or review date and if necessary, usage instructions should be clearly mentioned along with the download link of the document.
- iii. The date of publishing of the document along with the document title should clearly be stated
- iv. The language used in the document should be clearly stated.
- v. Document should not contain scanned image of text data, rather it should be text to PDF to ensure that they can be read by screen readers. The content in PDFs should be searchable.
- vi. Uploaded documents must be scanned for viruses and malware before being made available to the public.
- vii. The file names should be descriptive and follow a consistent naming convention, avoiding spaces and special characters. Instead, hyphens or underscores should be used.
- viii. There should be a clear specification of the maximum file size allowed for uploads. Websites should ensure that file sizes are optimized for quick downloads



**Quick Tip:**

Use File compression methods to reduce file size without compromising quality, making them more accessible for users.

*For more details, please refer to: GIGW 3.0 Section: 5.1.6*

## 11) Standards On Usage Of Content Delivery Network (CDN):

A content delivery network (CDN) refers to a geographically distributed group of servers which work together to provide fast delivery of Internet content.

A CDN allows for the quick transfer of assets needed for loading Internet content including HTML pages, JavaScript files, stylesheets, images, and videos.

**When to avoid the use of CDN:**

- When most of the website visitors are in the same geographic area. The main benefit of a CDN is lost if most of the website traffic is local, which is generally true for smaller, local businesses.
- When a CDN provider has no servers close to the website visitors. Select a CDN provider that has servers in the locations where most of the audience is, otherwise the advantage of the main benefit of CDNs is lost.
- When the website has low traffic. It's possible that a CDN could actually slow down the website. This applies to CDNs that use the "pull" model for caching files from the origin server, rather than the "push" model. With a pull CDN, the CDN is responsible for pulling the right content from the origin server to deliver to visitors.

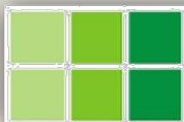
For Government website, NO external CDNs to be used, only if government of India / Goa has any CDNs available same can be used.

## 12) Security Standards:

Before deploying the website on the production server and completing the Go-Live phase, the website must be Tested and audited by a CERT-IN empanelled organization and security audit clearance certificate and report to be received from auditor. Website to be audited every year. Below are some of the common security standard patches that a website must implement before it is sent for audit

- CMS platform and plugins should be upto date
- The content should be delivered over HTTPS / SSL and latest TLS versions
- The password should be cryptic
- CMS default settings should be changed
- Enable CMS level security

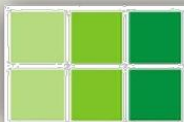




- Hide Apache web server and PHP versions
- Enforced HTTP to HTTPS redirect
- Prevent Cross Origin Resource Sharing (CORS, read the quick tip below)
- SSL Lab test grade should be A / A+, and Serpworx score should be 80 or above.

### Quick Tip on how to ensure website security:

1. **Keep Software And Plugins and Themes Up-To-Date:** Every day, there are countless websites compromised due to outdated software. Potential hackers and bots are scanning sites to attack. Crucial software's like the core Wordpress and Drupal softwares should be updated on timely manner. Themes and plugins used on the websites should be up to date. Other technologies being used such as jQuery, has to be of the latest version.
2. **Add HTTPS and an SSL Certificate:** To keep the website safe, a secure URL is needed. If the site visitors offer to send their private information, HTTPS is needed, not HTTP, to deliver it. SSL (Secure Sockets Layer) is another necessary site protocol. While procuring an SSL certificate, you must use a 2048-bit key to generate the CSR file. This transfers visitor's personal information between the website and the database. SSL encrypts information to prevent it from others reading it while in transit. SSL must be running on TLS version 1.3, versions 1.1 and 1.2 must be disabled. Disable 3DES cipher. This can be achieved for Apache httpd by setting: SSLCipherSuite HIGH:MEDIUM:!MD5:!RC4:!3DES
3. **Choose a Smart Passwords:** With there being so many websites, databases, and programs needing passwords, it is hard to keep track. A lot of people end up using the same password in all places, to remember their login information. But this is a significant security mistake. Create unique password for every new log in request. Come up with complicated, random, and difficult to guess passwords. Then, store them outside the website directory.
4. **Change Content Management System (CMS) Default Settings:** CMS settings can include adjusting control comments, user visibility, and permissions. A great example of a default setting change that should be made is 'file permissions.' Change the permissions to specify who can do what to a file. Each file has three permissions and a number that represents every permission:
  - i. 'Read' (4): View the file contents.
  - ii. 'Write' (2): Change the file contents.
  - iii. 'Execute' (1): Run the program file or script.Along with the default file permission settings, there are three user types:
  - Owner – Often, the creator of the file, but ownership can be changed. Only one user can be the owner at a time.
  - Group – Each file is assigned to a group. Users who are part of that specific group will gain access to the permissions of the group.
  - Public – Everyone else.
5. **Content Management System (CMS) Level Security:** After installing the intended CMS to be used for the website, make sure the following steps are integrated:
  - a) The version of the respective CMS (Wordpress / Drupal) should be hidden.
  - b) On log-in and registration page for administrator and users respectively, there needs to be a security Captcha having minimum six alpha-numeric characters.
  - c) User Enumeration should be disabled. Error messages on the login and registration pages should not show the username entered.

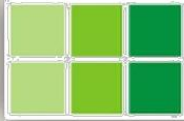


- d) There has to be a session limit of 30 minutes, so that an inactive logged-in user should be automatically logged-out after 30 minutes.
  - e) Restrict access to the Robots.txt file.
  - f) Restrict access to the xmprpc.php file.
6. **The Web Server Configuration Files:** Available in the root web directory. Web server configuration files permit you to administer server rules. This includes directives to improve website security.
- a) Apache web servers use the .htaccess file
  - b) Access to the main Apache server configurations file, should be disabled. Disable using any of the two Apache directives:
    - i. **ServerTokens:** The value of ServerTokens Full should be made "Prod"
    - ii. **ServerSignature:** The value of ServerSignature Full should be made "Off"
  - c) Similarly, the PHP version should be hidden. This can be achieved by changing the PHP directive from "expose\_php=On" to "expose\_php=Off".
7. **Implement HTTP Security Headers:** Security headers can be implemented inside the server configurations file, in the case of Apache web server, it is the .htaccess file, which is present in the root directory of the website. When a user visits a site through his/her browser, the server responds with HTTP Response Headers. These headers tell the browser how to behave during communication with the site. These headers mainly comprise of metadata. Types of security headers:
- a) **HTTP Strict Transport Security (HSTS):** If a site is equipped with HTTPS, the server forces the browser to communicate over secure HTTPS. This way, the possibility of an HTTP connection is eliminated entirely.
  - b) **Content Security Policy (CSP):** The HTTP Content Security Policy response header gives website admins a sense of control by giving them the authority to restrict the resources a user is allowed to load within site. In other words, you can whitelist the site's content sources.
  - c) **Cross Site Scripting Protection (X-XSS):** X-XSS header protects against Cross-Site Scripting attacks. This filter doesn't let the page load when it detects a cross-site scripting attack.
  - d) **X-Frame-Options:** An attacker fools a user into clicking something that isn't there. For example, a user might think that he's on the genuine website, but something else is running in the background. A user may reveal his/her confidential information in the process. X-Frame-Options help guard against these kinds of attacks. This is done by disabling the iframes present on the site. In other words, it doesn't let others embed the content.
  - e) **X-Content-Type-Options:** The X-Content-Type header offers a countermeasure against Multipurpose Internet Mail Extensions (MIME) sniffing. It instructs the browser to follow the MIME types indicated in the header. Used as a feature to discover an asset's file format, MIME sniffing can also be used to execute cross-site scripting attacks.

Security headers can be implemented inside the .htaccess file with the following syntax:

```
#security headers
<IfModule mod_headers.c>
#feature policy
```





```
Header set Feature-Policy "geolocation 'self'; vibrate 'none'"
#cookie flag
Header edit Set-Cookie ^(.*)$ $1;HttpOnly;Secure;SameSite=Strict;
#Content security policy
Header add Content-Security-Policy "default-src 'self' 'unsafe-eval' 'unsafe-inline' img-src * data:
blob: android-webview-video-poster.; fonts.googleapis.com *.gstatic.com
stackpath.bootstrapcdn.com translate.google.com translate.googleapis.com cdn.jsdelivr.net
google.com;"
#Referrer-Policy
Header set Referrer-Policy "same-origin"
#Permissions Policy
Header always set Permissions-Policy "geolocation=(),midi=(),sync-
xhr=(),microphone=(),camera=(),magnetometer=(),gyroscope=(),fullscreen=(self),payment=()"
#X-Content Type
Header set X-Content-Type-Options nosniff
#XSS-Protection
Header set X-XSS-Protection "1; mode=block"
Header unset X-Forwarded-Host
Header set X-Frame-Options "sameorigin"
Header set Strict-Transport-Security "max-age=63072000; includeSubDomains; preload"
</IfModule>
```

8. **Use of h2 and h3 Protocols instead of httpd/1.1:** Open the Apache server configuration file, "httpd.conf". Uncomment the following line:

```
"LoadModule http2_module modules/mod_http2.so"
```

After uncommenting the above line, add the following code in these two files: httpd.conf and https-ssl.conf just after the ports are being called: "Protocols h2 h2c http/1.1".

- i.) The above code will look like this in httpd.conf file:

```
"ServerName localhost:80
Protocols h2 h2c http/1.1"
```

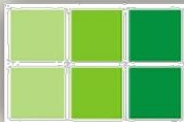
- ii.) The above code will look like this in httpd-ssl.conf file:

```
"<VirtualHost _default_:443>
Protocols h2 h2c http/1.1"
```

9. **Hide Apache Server Signature:** Apache's mod\_security modules is used to hide Apache server signature completely. It can be downloaded from the following

URL: "<https://www.apachelounge.com/download/>"

- i.) Click on mod\_security-2.9.7-win64-VS17.zip file like, download will start automatically.
- ii.) Extract the zip file contents, and open the mod\_security folder after file extraction.
- iii.) Copy the "mod\_security.so" file to the following directory: "[/xampp/apache/modules/](#)"
- iv.) Now go back to the mod\_security extracted folder, and copy the "yajl.dll" file and paste it in the following directory: "[/xampp/apache/bin/](#)"
- v.) Open the httpd.conf file and add the following line:



```
"LoadModule security2_module modules/mod_security2.so"
```

Vi.) Add the below code in the httpd.conf file at the end and save the file:

```
<IfModule security2_module>
    SecRuleEngine on
    ServerTokens Full
    SecServerSignature " "
</IfModule>
```

10. **Prevent Cross Origin Resource Sharing (CORS):** Cross-origin resource sharing (CORS) is a mechanism which enables controlled access to resources located outside of a given domain. It extends and adds flexibility to the same-origin policy. Add the below PHP code inside the functions.php file of the WordPress theme used:

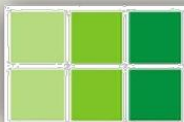
```
function initCors( $value ) {
    $origin_url = 'https://www.goa.gov.in/'; //ADD THE PRODUCTION URL HERE
    if (ENVIRONMENT === 'production') {
        $origin_url = 'https://www.goa.gov.in/'; //ADD THE PRODUCTION URL HERE
    }
    header( 'Access-Control-Allow-Origin: ' . $origin_url );
    header( 'Access-Control-Allow-Methods: GET' );
    header( 'Access-Control-Allow-Credentials: true' );
    return $value;
}

add_action( 'rest_api_init', function() {
    remove_filter( 'rest_pre_serve_request', 'rest_send_cors_headers' );
    add_filter( 'rest_pre_serve_request', initCors );
}, 15);
```

### 13)Website Performance Standards:

The website performance must be evaluated on the below parameters:

- I. Load Time: Website should load on a browser in 5 to 10 seconds on a 2G network and within 5 seconds on a 3G network
- II. Pagespeed Score should be atleast 90 on a desktop and mobile devices.
- III. GT Matrix grade should be A / A+.
- IV. All Javascript(JS) and cascading style sheets(CSS) must be minified.
- V. Images should be used in next-gen format.
- VI. Reduce the HTTP calls by combining the related JS and CSS in the same file.



## 14) Search Engine Optimization (SEO) Standards

Search engine optimization (SEO) is the science of getting pages to rank higher in search engines such as Google. Because search is one of the main ways in which people discover content online, ranking higher in search engines can lead to an increase in traffic to a website.

The website should compulsorily implement on-page SEO to ensure the website is listed on top in search engines. All the link on the website along with the content and page structuring should be developed considering search engine compliance.

The following components on the website should be compulsorily set to have best search results listing.

- I. **Title Tag**
- II. **Meta Description**
- III. **Canonical Tag**
- IV. **Alternate Text Tag**
- V. **Robots Meta Tag**
- VI. **Open Graph Meta Tags and Twitter Cards**
- VII. **Header Tags**
- VIII. **Alternate Image text**

Quick Tip:

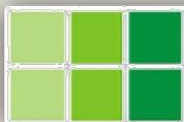
For more details, refer SEO compliance document

## 15) Website Management Standards:

- Configure website with Google analytics and Google search console, review monthly analytics. Observe how users are searching for the website and include those keywords in SEO.

Quick Tip

1. **Google Analytics:** Google Analytics is one of the most popular digital analytics software. It is Google's free web analytics service that allows you to analyze in-depth detail about the visitors on the website. It provides valuable insights that can help you to shape the success strategy of the business
2. **About Search Console:** Google Search Console is a free service offered by Google that helps you monitor, maintain, and troubleshoot the site's presence in Google Search results. You don't have to sign up for Search Console to be included in Google Search results, but Search Console helps you understand and improve how Google sees the site. Search Console offers tools and reports for the following actions:
  - i. Confirm that Google can find and crawl the website.
  - ii. Fix indexing problems and request re-indexing of new or updated content.



- iii. View Google Search traffic data for the website: how often the website appears in Google Search, which search queries show the website, how often searchers click through for those queries, and more.
- iv. Receive alerts when Google encounters indexing, spam, or other issues on the site.
- v. Show you which sites link to the website.
- vi. Troubleshoot issues for AMP, mobile usability, and other Search features.

- Ensure security standards are checked on monthly frequency
- Fix broken links on the website on monthly frequency.
- Maintain logs of the changes suggested in the website.
- Ensure personal and sensitive information as per IT Act is not being published on the website in text or PDF or image or any other attachment.

### Quick Tip:

Sensitive personal data or information: Sensitive personal data or information of a person means such personal information which consists of information relating to:-

- i. Passwords.
- ii. Financial information such as Bank account or credit card or debit card or other payment instrument details.
- iii. Physical, physiological and mental health condition.
- iv. Sexual orientation.
- v. Medical records and history
- vi. Biometric information.
- vii. Any detail relating to above clauses as provided to body corporate for providing service
- viii. Any of the information received under above clauses by body corporate for processing, stored or processed under lawful contract or otherwise.